

Plate Boundaries

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Divergent boundaries move apart, convergent boundaries move together, and transform boundaries slide past each other; each pattern is linked to characteristic landforms and geologic activity.
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Round 1 - Retrieve It

1. What happens at a divergent boundary?

2. What happens at a convergent boundary?

3. What happens at a transform boundary?

4. Which boundary types can produce earthquakes?

Round 2 - Sketch It

Draw three small boundary diagrams with arrows for divergent, convergent, and transform motion. Add one landform or event linked to each.

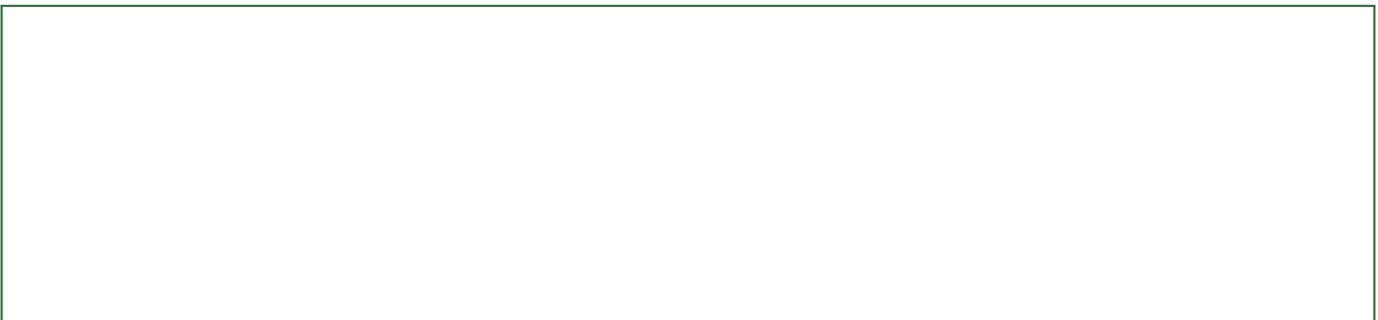


Plate Boundaries - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A mid-ocean ridge forms where seafloor spreads. Boundary type?

2. Two continental plates collide and build mountains. Boundary type?

3. Two plates slide past along a fault and produce earthquakes. Boundary type?

Round 4 - Reason from Evidence

Explain how the direction of plate motion helps you predict the kinds of geologic features that may form.

Plate Boundaries - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A
Articulate

Explain Plate Boundaries in your own words. Use one example that is not printed on the cue card.

C
Connect

Connect plate boundaries to the rock cycle. How can plate movement contribute to melting, uplift, or metamorphism?

E
Extend

A map shows a long ocean trench next to a volcanic arc. Infer the likely boundary type and explain your evidence.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Plate Boundaries

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Plates move apart.
2. Plates move toward each other.
3. Plates slide horizontally past each other.
4. All three can, though the settings and associated features differ.

Sketch It - look for

Divergent: arrows apart, ridges/rifts/new crust. Convergent: arrows together, mountains/trenches/volcanoes/subduction depending on plates. Transform: arrows sliding past, faulting/earthquakes.

Apply It

1. Divergent.
2. Convergent.
3. Transform.

Reason from Evidence - look for

Motion controls whether crust is created/separated, compressed/destroyed/thickened, or sheared sideways; those mechanisms produce different features.

ACE - Extend look-for

Convergent subduction boundary; trench and volcanic arc are common evidence of one plate sinking beneath another.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.